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Key Personnel Changes/Previous PI:	August 2016 report: Dr. Sunny Karnani of Clearsign Combustion Inc. was added as an Investigator.		
COI Name (Institution):	Karnani, Sunny Ph.D. (Clearsign Combustion Incorporated)		
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Task Description:	NOTE this is a successor agreement to "Applications of Electric Fields in Microgravity Combustion Science (NNX07AB55A)," per D. Stocker, NASA Glenn Research Center. This project at the University of California, Irvine explores the use of large electric fields in combustion for the purpose of improving the performance of energy conversion systems. As an actuator, electric fields can affect flame stability and soot formation, while, as a sensing device, the electrical response at saturation is an inherent flame characteristic. In addition to preparing for the proposed ACME E-FIELD experiments aboard the International Space Station (ISS), we have performed an evaluation of an experimental method of measuring ion number density in a flame using a Langmuir probe, studied the dynamic interaction between the flame and chemi-ions, analyzed the source of increased saturation ion currents found in I-V curves, used electric fields to control flames impinging on a surface, and begun to develop computational models to predict chemi-ion concentrations in flames. This report summarizes recent findings. Specific activities include: 1.) A study on the dynamic interaction between the flame and the electrical body force created by a flux of chem-ions. This was done in several ways, by analyzing the dynamic response of the gaseous flow field in response to a sudden application of an electric field, by comparing transient flame behavior using a high speed camera and the transient ion current collected when an electric field is turned on. 2.) A study of the interaction between the flame and the DC electric fields. This was done by studying the flame geometry to varies steady electric field strengths and analyzing I-V curves for different burner geometries. 3.) A study to construct a framework and approach for building computational models to predict ion concentrations in the flame, including the effects of ion driven winds. Different burner geometries were tested. Different ion chemistry models were used to test the current level of understanding of how ions are produced in the flames. Summaries of these findings appear in this report with the details of all the work appearing in the publications and conference proceedings identified at the end of this document. Task Significance: Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The control of combustion has the potential to improve efficiency and reduce emissions from burning fuels. Since high power density often requires combustion, these improvements will be important no matter what the fuel source. Electric fields acting on flames have the potential to aid in combustion control. In addition, electrical properties of flames can identify poor performing boiler flames that release poisonous carbon monoxide. Our studies show that a flame's electric signature can capture incipient quenching before dangerous emissions result. Understanding the links between electrical character and flame behavior may allow improved sensing of poor performing combustion systems.
	Experimental Activities: All ground-based experiments described were performed using a coflow burner that is dimensionally similar to the proposed ISS coflow burner but with a fuel jet matching the dimension of the ISS jet burner. In many experiments, the center fuel jet was slighting extruded above the top surface of the coflow burner to more closely approximate a jet burner. A variable high-tension power supply connected between the burner and downstream mesh electrode produces an applied electric field. Although the electric field distribution is deformed locally by the flame’s space charge, defining an applied field strength, E = V/H is a useful approximation when describing experimental results. Ion current is calculated using Ohm’s law and the measured potential drop across a shunt resistor placed between the electrically isolated burner and the building ground. An acrylic chamber shields the flame from room air currents. External electric fields have long been known to modify the geometry of the flame. It has been demonstrated since the midpoint of the 20th century that flames may deflect in the direction of an electrode with a negative electrical bias. This study seeks to understand the flame geometry behavior of a non-premixed, conical shaped flame that is stabilized on a coflow burner. The central tube is extruded 3 mm above the coflow surface. The electric field was initiated and the flame was given sufficient time to stabilize. Images were taken of the flame after the flame had reached its new electrically modified state. Flame images were taken under various electric field strengths. Abel inverted flames calculated from the images show the true contour of the chemiluminescence. Abel inverted flame contours were experimentally found under the influence of several electric field strengths. The general trend that was observed was that the flame height slightly decreases with an increasing applied field and the flame contour's radius also decreases with increasing field. It can also be seen that the general trend of brightness of the flame contour increases with increasing field strength. The normalized maximum light intensity in the brightest portion of the flame was found as a function of field strength. It was noticed that an initial peak is found at 1 kV/cm. Following this peak, a decrease was found up till 2 kV/cm followed by a large increase. This trend has also been seen in the ion current measurement. Finally, it was also observed that the high electric fields produced lifted flames which likely create a tribrachial flame region near the base of the flame. What effect this may be having on the light emission of the flame and ion current measurements (or vice versa) is yet to be determined. Current voltage curves (I-V curves) are used to characterize the strength of the electrical body force as a function of field strength. I-V curves can be used to determine what parameters create and control the electric field driven instabilities and secondary ionization mechanisms. This study also emphasizes the importance of taking ion current measurements when performing electric field combustion experiments. The purpose of this study is to determine a means of measuring the electrical body force through experimental parameters. Since the electrical body force is a result of not only an electric field but also a flux of flame ions, I-V curves may serve as a global indicator of the strength of the electric forcing. Typical I-V curves show that as the electric potential increases (from left to right) the ion current increases until a saturation point is reached. At saturation the ion current is balanced by the rate of ion production in the flame, therefore, no more ions can be extracted. Three burner configurations that were tested were a coflow burner, a coflow burner with the central fuel tube extruded 3 mm above the top of the coflow section, and a circular jet nozzle.

Task Progress:

The I-V curves were obtained by gradually increasing the electrode potential and holding each voltage for 2 seconds. In the coflow burner with an extruded fuel tube, the saturation current began to increase close to 1.5-2 kV/cm. For the jet nozzle, the saturation current appeared constant after saturation, however, large error bars appeared which indicated large deviations of the measured values from averaged values. This occurred because the jet flow/flame became unstable soon after the saturation.

The increase beyond saturation ion currents has been measured before but not well explained. Two hypotheses may explain the increase: (1) secondary ionization may be more prevalent due to large field strengths; electrical breakdown of the gas is unlikely due to the relatively low field strengths used in these experiments. However, local breakdown due to corona discharge may provide a source of secondary ionization. Another source of secondary ionization could come from thermi-ionization from the hot surfaces. (2) flame dynamics may also provide an explanation to the increase in saturation ion current; that is, a strong ion wind may provide the flame with an environment in favor of enhanced chemi-ionization reaction rates. The hydrogen flame test presents the case of no chemi-ionization reaction but with a hot burner/electrode. The results of this test were compared to the methane tests. With a hydrogen flame stabilized on a flat coflow burner, increases to the ion current measurements were not observed. This suggests that secondary ionization is not a factor in these experiments. However, an ion current is measured with the hydrogen flame stabilized on the coflow burner with an extruded central fuel tube above 3.5-4 kV/cm. However, It should be noted that the methane equivalent tests show an increase in saturation ion current much before these field strengths, and thus, it may be concluded that the increase in saturation ion currents are not initially affected by secondary ionization.

The deviation from the averaged methane ion current values shows an interesting correlation when compared to the hydrogen ion current measurements. In nearly all the data points the error bars are negligible because the ion current measurements were very stable under a steady DC electric field. However, above 3 kV/cm slight deviations from the averaged values were measured indicating a small instability in the ion current measurements. It was also observed that the methane flame became highly unstable and eventually became extinguished at 4 kV/cm. It was also observed that the ion current measurements from the hydrogen flame are occurring at these very high field strengths. The same observation, however more dramatic, was made in the jet burner I-V curves. These observations must suggest that the effects of ionization are not the source of increasing saturation ion current measurements but are the source in the flame/ion current instability. In order to understand more clearly these interactions, further experimentation is needed. However, the current hypothesis focuses around an ion wind affecting the flame/burner interaction. That is, the large ion winds that were seen to be lifting the flame in the last section could be reducing the heat transfer to the burner, thus allowing for higher chemi-ionization reactions to occur. Also, lifted flames are known to produce partially premixed edge flames near the flame base. Premixed flames have been known from the literature to produce higher ion concentrations than non-premixed flames. More studies will be performed in the future to better understand these results.

A schlieren system was built in the lab to study the dynamic flow effect of instantaneously switching on an electric field. At t=0, a negatively biased electric potential is applied to the downstream mesh electrode. The potential is reached within 1 ms after being triggered. When the high voltage electric potential is triggered and applied to the mesh electrode, a pinching like effect is observed in the schlieren column close to the fuel tube exit. The pinching effect propagates along the length of the schlieren column in a wave-like motion. The initial wave is followed by 1-2 more secondary waves that follow until the entire schlieren column re-stabilizes into a thinner column than prior to the external field. The entire process lasts approximately 100 ms for re-stabilization to be achieved.

This analysis was used to determine where the wave was in time. The same process was used but this time using more heights. This was done for multiple field strengths for a 3.5 cm electrode spacing and 27 sccm fuel flow rate. It should also be noted that no modification to the schlieren image or flame behavior could be observed with field strengths less that 0.6 kV/cm. This would imply that the ion flux is too small to produce a significant body force in the gas. A first order polynomial fit was used to determine the wave velocity. The velocities were plotted as a function of velocity. Because the velocity of this thermal wave is related closely to the convection of the gas, it is implied that the velocity of the gas downstream of the flame could be accelerated by an order of magnitude by high external fields. It was also seen from that comparing a 1.1 kV/cm to a 2.7 kV/cm field strength an ion current increased from 1.7 to 2 μA and subsequently produced an increase in thermal wave velocity from 373 to 700 cm/s. That is, a 17\% increase above the saturation ion current produced a 90% in the thermal wave velocity. One thing that could explain this is the increase in field strength. Future studies will take a closer look at local current densities within the electrode gap to understand better the distribution of migrating charges at various field strengths.

Numerical Simulation Activities: The aim of this study is to better understand, through numerical simulations, the effects of chemi-ionization and electric fields on a non-premixed coflow flame that impinges on a metallic plate; this configuration is very useful in order to investigate both how electric fields can be used to reduce carbon monoxide emissions and how they affect the heat flux on a solid surface. In order to analyze how chemi-ionization and electric fields influence the fluid dynamics and the chemistry of a flame in this configuration, numerical simulations have been performed using OpenFOAM. The validation of the numerical model has been performed by comparing numerical results to experimental results; these comparisons show that, taking into account some simplifying assumptions (e.g., axialsymmetric geometry, absence of radiation heat losses, unitary Lewis and Schmidt number) introduced in the model, results agree well with literature findings.

Study Case 1-- The reduced ions mechanism used in the study of the non-premixed coflow flame that impinges on a metallic plate was developed using the simplest reduction concept of combining what looked to be the most important ion reactions with the most important related neutral molecule reactions in combustion. It is a mechanism that has not been validated in other systems so it represents more of an exploration of the level of computation that might be feasible for the entire electrically activated flame system. The plate itself acts as the negative electrode, while the burner is the positive electrode. A negative potential and a zero potential will be assigned to the plate and to the exit area of the burner respectively. The effects of the self-induced electric field can be summarized in this way: (a) the peak temperature leads to a 5% overestimation of the experimental results; (b) the electric potential (and subsequently the electric field) is established only in the flame zone, because it is only here that charges are created; (c) the bulk velocity is slightly increased (with respect to the case when the Poisson’s equation is not modeled in the solver) by the inclusion of the charged species in creating an electric field; (d) the mass fraction of CH3 increases, and this reflects in an increase of the mass fraction of CH; this enhances the chemi-ionization process, subsequently the mass fraction of key ions also increase. This happens because now the solver takes into account the charge accumulation in the flame zone; this means that the weak self-induced electric field facilitates the charge generation in the flames zone. And the charges created here tend to move away from each other only because of self-repulsion, since no electric field has been applied on the plate.

The preliminary simulations performed in the current study showed to have good comparison with experiments and very good predictions of the behavior of impinging flames in general. The results also show good agreement with peak temperatures (error estimation under 10%), and the temperature profiles are very similar to those obtained with the schlieren technique. In addition, the mass fraction of important species like CO and OH match qualitatively and quantitatively; and the mass fractions of charged species also match either with those expected from literature and with those obtained from other numerical simulations.

Study Case 2 -- In a second case, chemiluminescence of the flame is also taken into account. Excited species are important as an indicator for many flame properties including reaction zone position. It would be valuable, therefore, to add reactions involved with important excited species, such as excited hydroxyl, OH*, and methylidyne, CH*. At the moment, there is no published model for non-premixed co-flow methane/air flames that contains both chemi-ions and excited species. For this case, a new mechanism that combines the mechanisms from two known chemical mechanisms from the literature (1 methane-air neutral chemical mechanism and the other a Chemilumenscence mechanism) and a selection of reactions from GRI-Mech 3.0 that were important for the formation and consumption of ions and excited species is used in OpenFoam simulations without the metallic plate placed above the burner. Excited species were taken into account due to chemiluminescence of OH* and CH* excited radicals.

Hence, a new compact chemical kinetic mechanism of 73 reactions and 33 species (neutral molecules, chemi-ions, electrons, and excited species) is used for the study of the non-premixed burner configuration without metallic plate placed above the burner. In the simulation for the burner configuration without a metallic plate placed above the burner, the same user-defined solver used for the Study Case 1 was used; as for the Study Case 1, a zero potential has been set on the plate and the burner walls. A summary of the conclusions are described below: (a) it has been demonstrated that the zones with high concentration of OH (> 0.0017) coincide with the zones where OH* appears; (b) Results for CH and CH* tend to do the same but in this case, in some regions such as the top of the flame where there is less concentration of CH, there is still a considerable amount of CH*. Further research has to be done to predict better results for CH and CH* to match with the experimental ones; (c) zones with high concentration of HCO+ in the flame have low concentration of CH. Because there is a strong link between CH and CH*, also a high concentration of HCO+ would mean a low concentration of CH*, and vice versa. On the other hand, some other variable is needed to explain the relation between CH* and HCO+. That variable may be the oxygen concentration, which is also important for the chemo-ionization reaction; (d) generally, the highest concentration of H3O+ will be in the regions where the concentration of HCO+ is lower. However, because the reaction of production of H3O+ requires 4 times more energy of activation than the reaction for the production of HCO+, we might expect some regions with high concentration of HCO+ and also high concentration of H3O+; (e) H3O+ is transported far away. HCO+ is produced but is very short lived and reacts very quickly into H3O+. H3O+ has a much longer life time which explains how it is able to travel further outside the flame and have much higher concentrations than other ions.

Study Case 3 -- In order to delve deeper into the chemical reactions that could be giving rise to the results found in the CFD simulation of the small diffusion flame, and in order to separate effects that could be due to transport and mixing, the 1-D counter-flow configuration in CHEMKIN® was employed.

Two reaction mechanisms, GRI-Mech 3.0 v.2 and GRI-Mech 3.0 v.3., are compared. Both are based on the GRI-Mech 3.0 for the neutral species, including the reactions for the excited species. However, the GRI-Mech 3.0 v.3. also includes a well-known ion model from the literature. These were compared to the new mechanism (NC Model) described in Study Case 2.

For the main neutral species and temperature profiles, the results between the three versions of the GRI-Mech 3.0 produced no variation. Hence, the results for neutral species and temperature profile for the GRI-Mech 3.0 are compared with the results for the NC Model.

The temperature profiles reveal a maximum difference of less than 3%. Also, the peak of temperature (2000K and 2040K for GRI-Mech 3.0 and NC Model, respectively) is located at the same axial distance (0.755 m and 0.769 m for GRI-Mech 3.0 and NC Model, respectively), again with less than 3% difference. The neutral species profiles shapes are similar in both cases.

The temperature profiles for all models are similar and the location of the peak of maximum concentration of CH the same for all models studied. However, the NC Model shows concentrations of CH that are 4 orders of magnitude below the ones predicted by the GRI-Mech 3.0 models, and the NC Model shows an increase of the concentration of CH radical as soon as the temperature rises, while the increment of CH concentration begins 0.2 mm before the peak of maximum temperature.

Similar to the CH, the peak of maximum concentration of CH* is located at the same axial location for all reaction models studied. However, the width of the peak is slightly different. For the NC Model, the CH* mole fraction rises around 6.4 mm of axial distance while for the GRI-Mech 3.0 based models it is at 6.6 mm.

For the main ionic species, H3O+ and HCO+, a comparison was also made. In this case, the only mechanism based on the GRI-Mech 3.0 that contains ionic species is the v.3. of the model. The hydronium ion profiles present significant differences depending on the reaction model used. Concentrations of hydronium ion using the NC Model are two orders of magnitude below the ones obtained using GRI-Mech 3.0 v3. Even though the peak of maximum hydronium concentration for both cases is around 6.65 mm (axial distance), the concentration of H3O+ increases sharply for the GRI-Mech 3.0 v3 around 6.65 mm while it increases gradually for the NC Model since at axial distance of 4.25 mm.

The formyl cation profiles show fewer differences between mechanisms. The concentration of this ion for the NC Model presents a larger width than the one for the GRI-Mech 3.0. v.3. However, the peak of maximum concentration is located at the same axial distance for both mechanisms.

In all cases, the species profiles have larger widths for the NC model compared to the GRI-Mech 3.0. v.3. It is readily shown that that the difference between widths increases along the chemical pathway from CH to HCO+ to H3O+ . This is most likely due to errors that are generated during the prediction of CH that propagate down this chemical pathway into the prediction of chemi-ions. Further work is needed to determine the source of these errors. However, it seems that the C2H and O species are closely related with these differences due to the fact that they are the precursors of CH.

Study Case 4 -- The experience described above suggests that it will be difficult to couple the small concentration ions with the overall flow field in a numerical simulation. To have a better chance at accomplishing this coupling, we are

	exploring a highly adaptive mesh refinement approach pioneered at Lawrence Berkeley National Laboratory (LBNL). The code under development at LBNL uses BoxLib, a software library containing the functionalities (written either in C++ and Fortran90) required to perform adaptive mesh refinement (AMR); the basic idea of this technique is that the governing equations can be solved, at each time step, on a multilevel grid: at each level, the mesh is refined only where it is needed (typically where high gradients occur), and the solution is recomputed only in these zones. This allows to refine the solution without having to recompute it on the entire domain, which saves time and computational costs. BoxLib can solve problems with simple geometry (i.e., boxes), in 2D and 3D, and in either Cartesian and cylindrical coordinates; it supports hybrid MPI and OpenMP parallel programming. Results were obtained with the existing LMC code for a premixed and diffusion flame respectively; in both cases, it can be noticed that the different levels of refinement are located primarily close to the flame zone. The temperature profile for a premixed flame obtained with the existing 1D code was found without the AMR. The current study is focused on understanding the effects of an electric field on hydrocarbon flames; to take into account these effects, the Poisson's Equation has to be added to the set of conservation equations (mass, momentum, energy, and species) already implemented in the low Mach number combustion (LMC) code. The first case under development consists in modifying an existing 1D code, which works for the case of unitary Lewis number and without AMR; this is the simplest case scenario. Once this code will be ready, the next steps will be: include the AMR into the 1D code (including non unitary Le), and then modify the 2D code. NOTE: Continues as "ACME: EFIELD – ELECTRIC FIELD EFFECTS ON LAMINAR DIFFUSION FLAMES," grant NNX17AC51A with the same Principal Investigator Dr. Derek Dunn-Rankin.

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Abstracts for Journals and Proceedings	Tinajero J, Alves A, Bernard G, Autef L, Dunn-Rankin D. "Using Voltage Current Curves To Characterize Electric Field Effects On Flames For Different Burner/Mesh Electrode Configurations." Oral presentation to be at 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Lopez-Camara C-F, Tinajero J, Ricchiuti V, Dunn-Rankin D. "Numerical Simulation of Flames under the Influence of Electric Field." Oral presentation to be at 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
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